Project Guidelines

CS-GY 6763: Algorithmic Machine Learning and Data Science

Fall 2026

Goal of the Project

The goal of this project is to **demonstrate a deep understanding of an important theoretical result** and communicate that understanding in a really well-written manuscript. Think of your manuscript as the first thing you would send to someone who wants to learn the result. Ideally, it should become a reference that other students and researchers find useful, perhaps even the canonical place to start learning about the result.

With AI tools available to help explore ideas and work through proofs, I want this project to put particular emphasis on understanding. You should know what the theorem says, why it matters, and why it is true. Proving a new theorem would be amazing as well, but **a new result is not required for an excellent project**. A clear explanation of an important result is itself a valuable contribution.

Choosing a Result

Work in **groups of two students**. The list of suggested papers provides recent results that can serve as starting points. You are also welcome to choose your favorite major theoretical breakthrough related to the course. The list is meant to help you find a topic, not to restrict your choices.

Start with one main paper, and expect to read other papers to understand its context and the tools it uses. Choose a result you find exciting enough to spend time understanding. If the full theorem is too broad or technically demanding, it is fine to focus on a central special case that captures the main ideas. Talk to the TAs or me if you would like help choosing a paper or deciding on an appropriate scope.

What Your Manuscript Should Do

Write for a reader who is comfortable with the course material but has not read your chosen paper. Your manuscript should give that reader a coherent route from the problem to the result and its proof.

1. **Explain the problem and why it matters.** What question does the paper answer? What was known before, and what made this question difficult? Give enough context for the reader to appreciate the result.

2. **State the result precisely.** Introduce the definitions and assumptions needed to understand the theorem. Explain what its guarantee means and how it improves on what was previously known. Examples can help make an abstract statement concrete.

3. **Explain why the result is true.** Give a roadmap of the proof, identify the main ideas, and then work through the central argument carefully. Help the reader understand why each step is useful. Fill in details that the original paper leaves implicit when those details are important for understanding the proof.

4. **Choose a scope you can explain well.** You do not need to reproduce every technical detail of a long paper. You can develop a simplified proof or prove a meaningful special case in full detail. Be explicit about which statements you prove and which you use from other sources, and explain how your treatment relates to the full result.

5. **Make the exposition your own.** Use your own organization, explanations, examples, and figures to make the ideas easier to learn. Cite the original result and any other sources you use. Give credit for proof ideas and distinguish any new observations of your own. Every member of the group should be able to explain and justify the arguments in the manuscript.

Paper-Choice Checkpoint: Friday, November 6, 2026

Submit a short proposal that includes:

- The names of both group members.
- The title, authors, and a link to the paper you have chosen.
- The result you plan to explain, and why you chose it. What makes it important or interesting to you?
- Your intended scope: the full result or a particular special case, and the main ideas you hope to help the reader understand.

You are not expected to understand the full proof at this checkpoint. The goal is to make a thoughtful choice and give us enough information to help you refine the project.

Final Manuscript

The final deliverable is your manuscript. The final submission date will be announced separately. We will evaluate it on **mathematical understanding, correctness, and quality of exposition**. Choose a scope that lets you explain the central ideas in depth, and leave time to revise.